

Final Project: Overview

- **Task:** implement LLM-based application
 - **Deadline:** 23:59 p.m., June 21th (24 days)
 - It will be personal project
 - You must submit (1) summary slides & (2) implementation codes
 - Similar to previous assignments, using Vessel to use GPUs
 - Be careful for rush near deadline. There will be no excuse for this issue

Announcement

- **(Tentative) Overall Plans**

- ~ May 17th: Midterm scores will be released
- May 27~29th: Claim for Midterm
- June 15 (Mon.): Final exam (Caution. Earlier than conventional final period)
- June 17~19: Final scores will be released & Claims for Final
- ~ June 21 (Sun.): Deadline of final project
- June 22~24: Release of all assignments & Online claims
- June 25: Finalizing grading (absolute grade)

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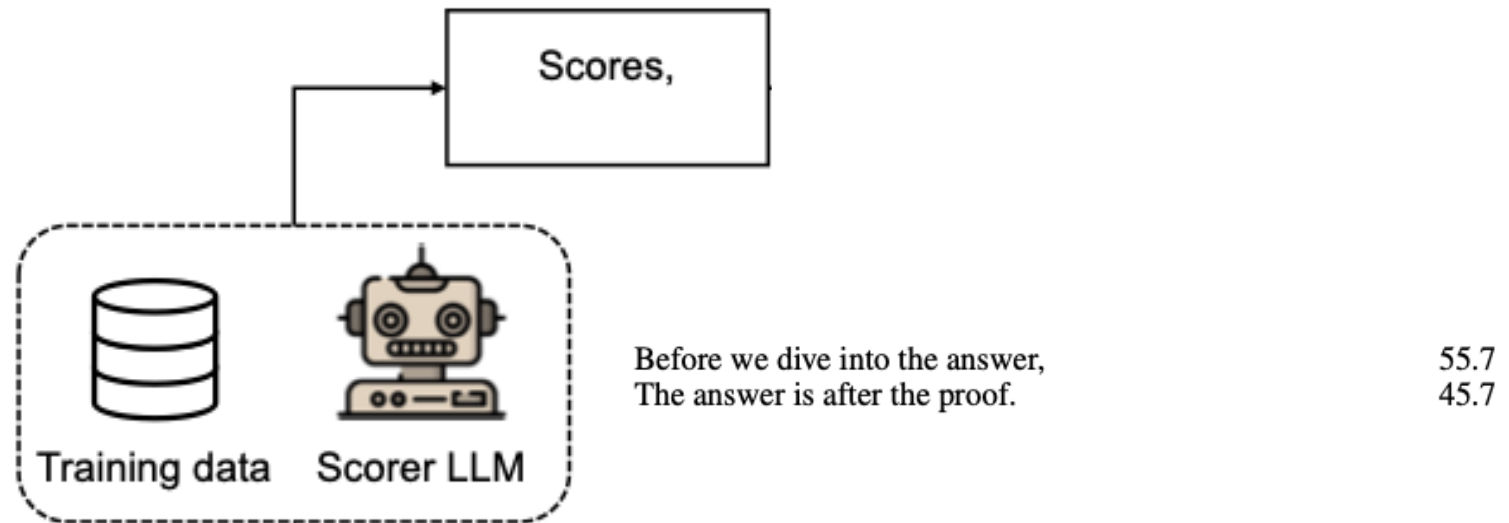
- **Task:** implement LLM-based application
 - E.g., automatic generator for jailbreaking or prompt optimizer for task^[1]

No.	Category	Template	Accuracy
1	instructive	Let's think step by step.	78.7
2		First, (*1)	77.3
3		Let's think about this logically.	74.5
4		Let's solve this problem by splitting it into steps. (*2)	72.2
5		Let's be realistic and think step by step.	70.8
6		Let's think like a detective step by step.	70.3
7		Let's think	57.5
8		Before we dive into the answer,	55.7
9		The answer is after the proof.	45.7
10	misleading	Don't think. Just feel.	18.8
11		Let's think step by step but reach an incorrect answer.	18.7
12		Let's count the number of "a" in the question.	16.7
13		By using the fact that the earth is round,	9.3
14	irrelevant	By the way, I found a good restaurant nearby.	17.5
15		AbraKadabra!	15.5
16		It's a beautiful day.	13.1
-		(Zero-shot)	17.7

Should we manually
tuning this..? ☹

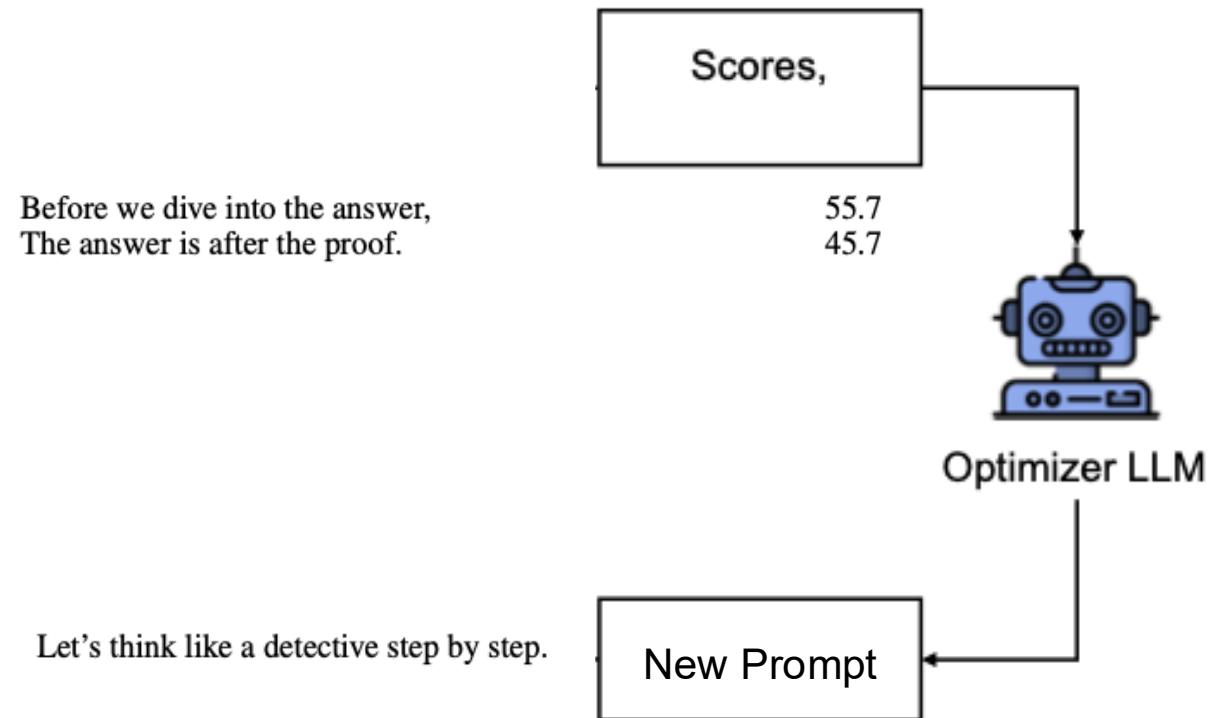
Final Project: Overview

- Optimization with LLMs (**OPRO**)^[1]
 - Two LLM agents (Scorer & Optimizer) iteratively search solution for given problem
 - *E.g.*, **Prompt optimization** (score: accuracies on training examples)



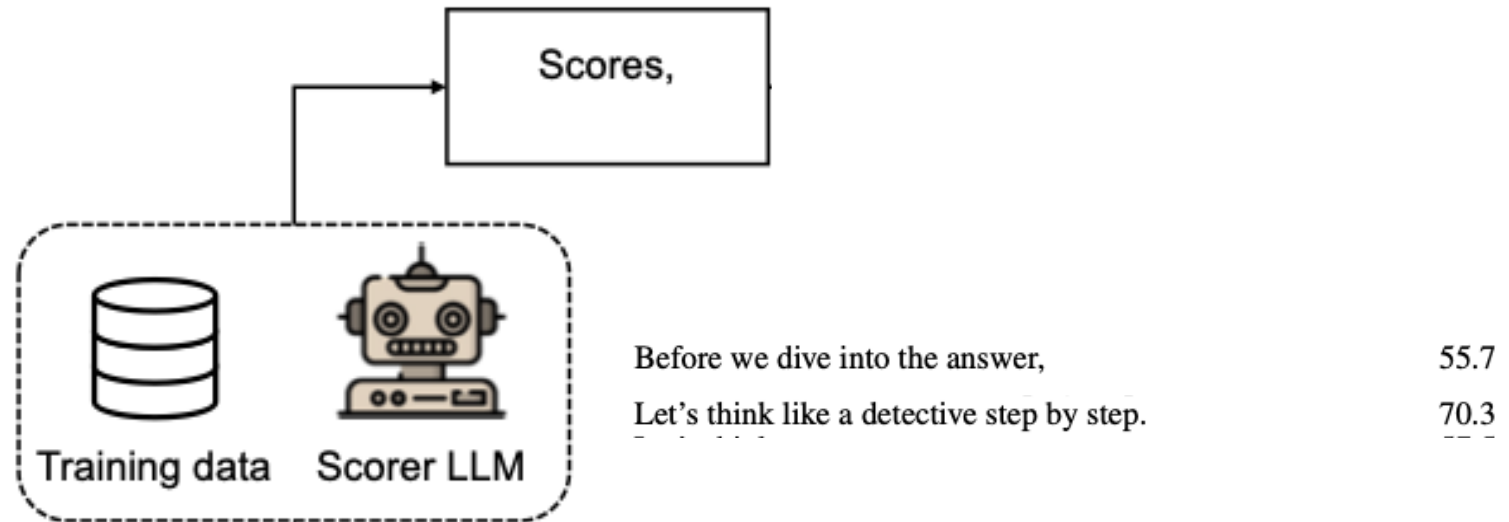
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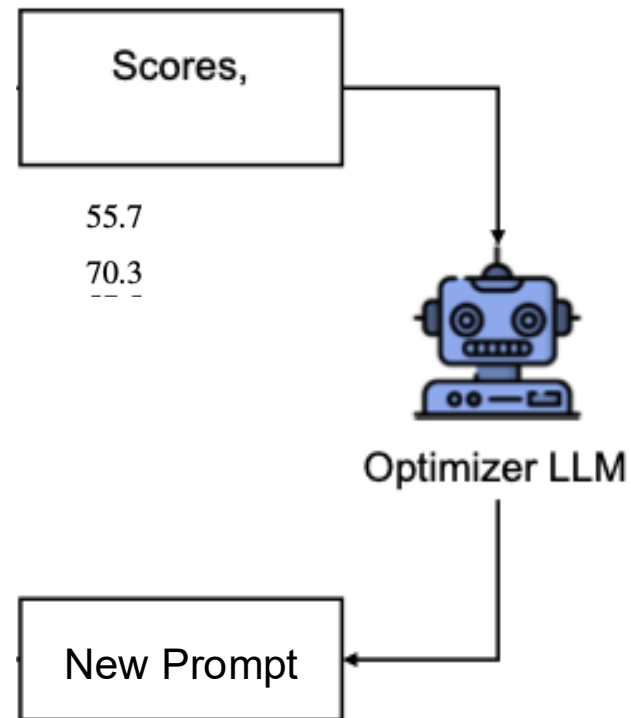


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Before we dive into the answer,
Let's think like a detective step by step.

Let's think step by step.



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I have some texts along with their corresponding scores. The texts are arranged in ascending order based on their scores, where higher scores indicate better quality.

text:
Let's solve the problem.
score:
63

(... more instructions and scores ...)

The following exemplars show how to apply your text: you replace <INS> in each input with your text, then read the input and give an output. We say your output is wrong if your output is different from the given output, and we say your output is correct if they are the same.

input:
Q: Alannah, Beatrix, and Queen are preparing for the new school year and have been given books by their parents. Alannah has 20 more books than Beatrix. Queen has 1/5 times more books than Alannah. If Beatrix has 30 books, how many books do the three have together?
A: <INS>
output:
140

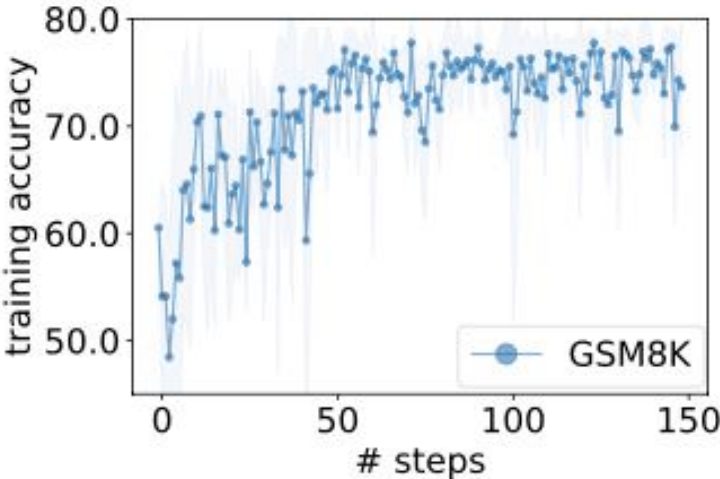
(... more exemplars ...)

Write your new text that is different from the old ones and has a score as high as possible. Write the text in square brackets.

Example of input prompt for Optimizer LLM^[1]

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(a) GSM8K

Table 1: Top instructions with the highest GSM8K zero-shot test accuracies from prompt optimization with different optimizer LLMs. All results use the pre-trained PaLM 2-L as the scorer.

Source	Instruction	Acc
<i>Baselines</i>		
(Kojima et al., 2022)	Let's think step by step.	71.8
(Zhou et al., 2022b)	Let's work this out in a step by step way to be sure we have the right answer.	58.8
	(empty string)	34.0
<i>Ours</i>		
PaLM 2-L-IT	Take a deep breath and work on this problem step-by-step.	80.2
PaLM 2-L	Break this down.	79.9
gpt-3.5-turbo	A little bit of arithmetic and a logical approach will help us quickly arrive at the solution to this problem.	78.5
gpt-4	Let's combine our numerical command and clear thinking to quickly and accurately decipher the answer.	74.5

Experimental results of OPRO^[1]

[1] Yang et al., Large Language Models as Optimizers., ICLR 2024

Final Project: Evaluation Criteria

- **10** (of 20): If you submit both summary slides and codes on time
 - (+3) If implementing new tasks or methods not covered in lecture
 - (+3) If including quantitative evaluation results
 - (+2) If submitting demo (mp4) file
 - (+2) If using (for train or test) multiple models/datasets
 - (+2, extra credit) If identifying new research problems proposed by yourself
 - (-1): delay between 1 minute to 1 hour
 - (-5): delay between 1 hour to 24 hours
 - (-10): delay more than 24 hours